# The GHGA Metadata Model: A framework for National Data Sharing

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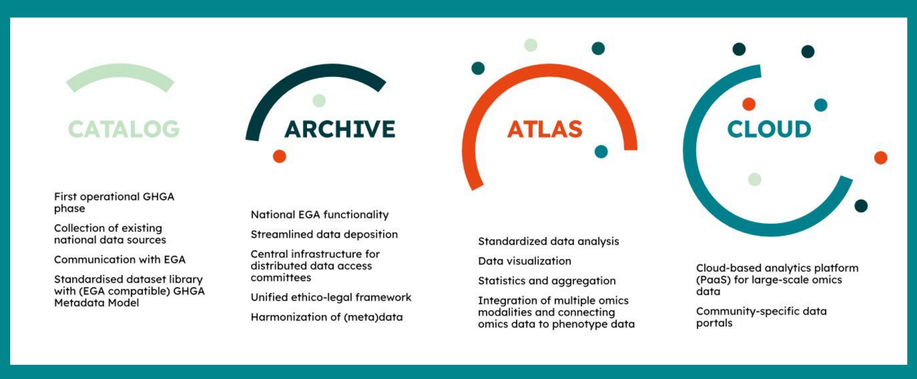
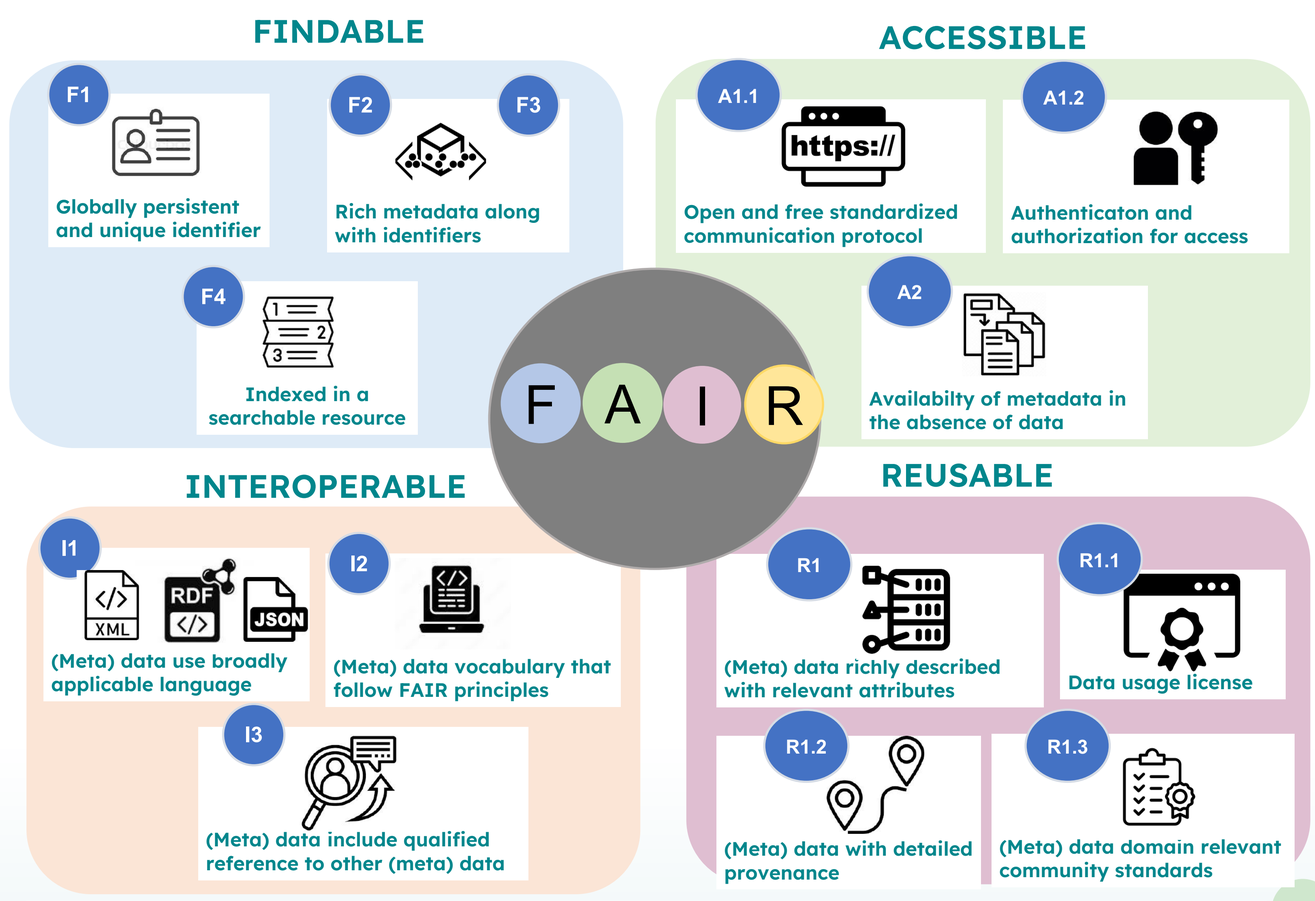
## About GHGA

The German Human Genome Phenome Archive (GHGA) is an initiative funded by the National Research Data Infrastructure (NFDI) since 2020 to build a national federated infrastructure to store and share human omics data.




## Mission of GHGA

To protect genomic data
To make genomic research easier
To provide international visibility to German Research

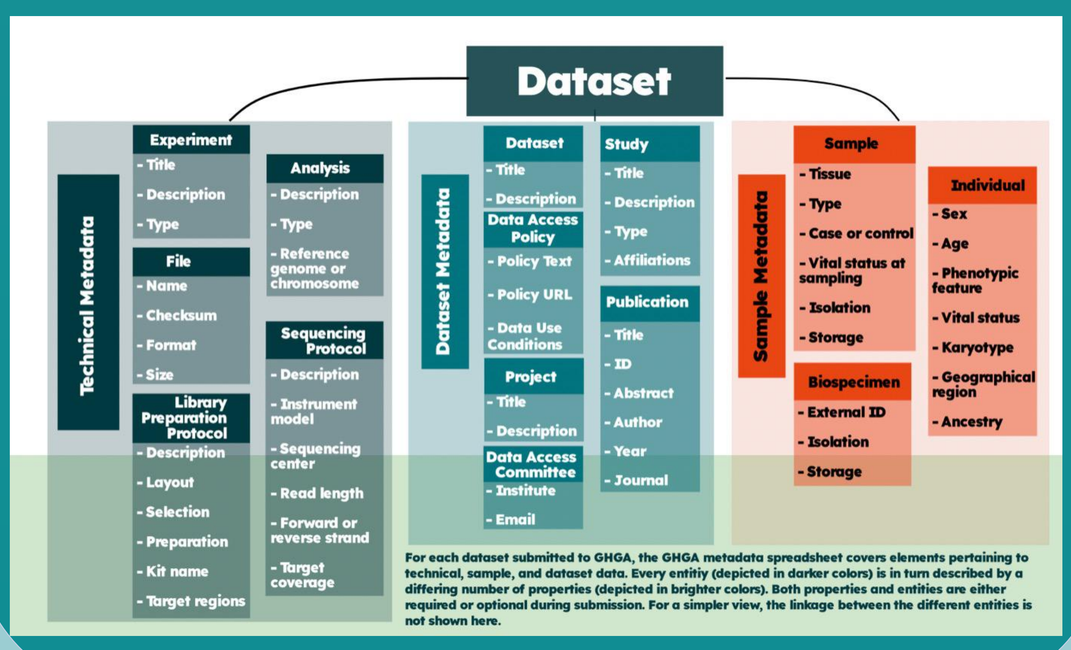



### Community Standards and Ontologies in use

Global Alliance for Genomics & Health — Collaborate. Innovate. Accelerate. | NIH NATIONAL CANCER INSTITUTE GDC Data Portal | human phenotype ontology | Data Use Ontology | ICD-10 | mondo | genomic STANDARDS consortium | DublinCore | NCIthesaurus | EBISPOT/ancestro | SNOMED CT

## GHGA Metadata Model

The goal of the metadata model is to help our communities to enrich their submitted genomic metadata as well as to retreive data of interest. The schema is modeled using LinkML and exists as a Yet Another Modeling Language (YAML).




For each dataset submitted to GHGA, the GHGA metadata spreadsheet covers elements pertaining to technical, sample, and dataset data. Every entitiy (depicted in darker colors) is in turn described by a differing number of properties (depicted in brighter colors). Both properties and entities are either required or optional during submission. For a simpler view, the linkage between the different entities is not shown here.

## GHGA Submission Process

### Extract from submission sheet

| CORE SUBMISSION SHEET | | |
|---|---|---|
| EXPERIMENT | | |
| | Captured? | Comments/Remarks |
| **submitter_experiment_alias** | | |
| experiment_title | | |
| **experiment_description** | | |
| experiment_type | | |
| experiment_biological_replicates | | |
| experiment_technical_replicates | | |
| STUDY | | |
| **submitter_study_alias** | | |
| **study_title** | | |
| **study_description** | | |
| **study_type** | | |
| **study_affiliations** | | |
| study_attributes | | |
| PROJECT | | |
| **submitter_project_alias** | | |
| **project_title** | | |
| **project_description** | | |
| project_attributes | | |
| PUBLICATION | | |
| publication_title | | |
| **publication_id** | | |
| publication_xref | | |

| EXPERIMENTAL SUBMISSION SHEET | | |
|---|---|---|
| EXPERIMENT AND FILE IDENTIFIER | | |
| | Captured? | Comments/Remarks |
| **submitter_experiment_alias** | | |
| **submitter_file_alias** | | |
| LIBRARY PREPARATION PROTOCOL METADATA | | |
| **submitter_libprep_alias** | | |
| **libprep_library_description** | | |
| libprep_library_name | | |
| **libprep_library_layout** | | |
| **libprep_library_type** | | |
| **libprep_library_selection** | | |
| **libprep_library_preparation** | | |
| libprep_kit_retail_name | | |
| libprep_kit_manufacturer | | |
| libprep_target_regions | | |
| libprep_rnaseq_strandedness | | |
| libprep_primer | | |
| libprep_end_bias | | |
| libprep_attributes | | |
| SEQUENCING PROTOCOL METADATA | | |
| **submitter_sequencing_alias** | | |
| **seq_description** | | |
| seq_type | | |
| **seq_instrument_model** | | |

| SAMPLE DATA SUBMISSION SHEET | | |
|---|---|---|
| SAMPLE | | |
| | Captured? | Comments/Remarks |
| **submitter_sample_alias** | | |
| **submitter_experiment_alias** | | |
| **sample_name** | | |
| **sample_tissue** | | |
| **sample_description** | | |
| sample_type | | |
| sample_case_control_status | | |
| sample_vital_status_at_sampling | | |
| sample_isolation | | |
| sample_storage | | |
| sample_index_sequence | | |
| sample_lane_number | | |
| sample_xref | | |
| sample_attributes | | |
| BIOSPECIMEN | | |
| **submitter_biospecimen_alias** | | |
| biospecimen_external_id | | |
| biospecimen_name | | |
| biospecimen_type | | |
| biospecimen_description | | |
| biospecimen_isolation | | |
| biospecimen_storage | | |

### Guided Submission

| Disclosure Considerations | PROPERTY NAME | # Experiment and File Identifier (Mandatory) | | |
|---|---|---|---|---|
| | | submitter_experiment_alias | submitter_file_alias | submitter_libprep_alias |
| Aliases, Titles and Descriptions should not be based on the personal information of the Data Subject(s) and should not contain information that could aide to reidentify Data Subject(s). For example, generating an alias through combining someone's initials and birth date. In circumstances where the project is based on data regarding a single Data Subject, the project alias could serve as a psuedonym for that person. As such, you should take care to secure any file that links an project alias to the personal information of the Data Subject. | PROPERTY DESCRIPTION | An alias specified by the data submitter to uniquely identify and link the experiment metadata. Also referenced in the core submission sheet to link both metadata objects. | An alias specified by the data submitter to uniquely identify and link the file metadata. Also referenced in the file submission sheet to link both metadata objects. This MUST be the same as in the File_Submission_Sheet. | An alias specified by the data submitter to uniquely identify and link the library preparation metadata to an experiment. |
| | DATA TYPE | string | string list (semicolon separated) | string |
| | MANDATORY/OPTIONAL | mandatory | mandatory | mandatory |
| | HOW TO | The alias is used for internal data linkage. **The experiment alias will be shown publicly as experiment ID!** | **NOTE: The aliases are for internal linkage and data owners** | **NOTE: The aliases are for internal linkage and data owners** |

### Submission Information

| GHGA | Type | Version |
|---|---|---|
| | CORE | 0.9.1 |
| Contact | <GHGA Helpdesk EMAIL><br><GHGA Metadata Team EMAIL> | |
| Additional Information | | |

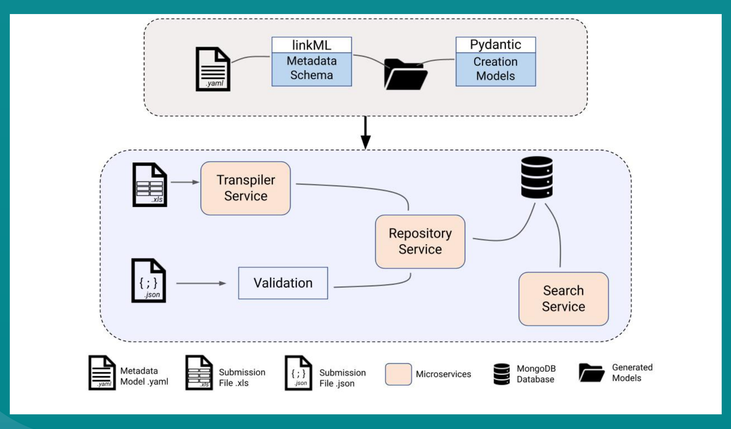
| Entity | Study | File | Library Preparation | |
|---|---|---|---|---|
| Property | study_type | file_format | libprep_library_layout | libprep_library_type |
| Permissible Values: | Cancer Genomics | agp | SE | WGS |

Core -> File -> Experiment -> Sample -> Analysis -> Dataset

## GHGA Post-Submission Process

The Transpiler Srvice transforms submission excel spreadsheets into a validated JSON representation

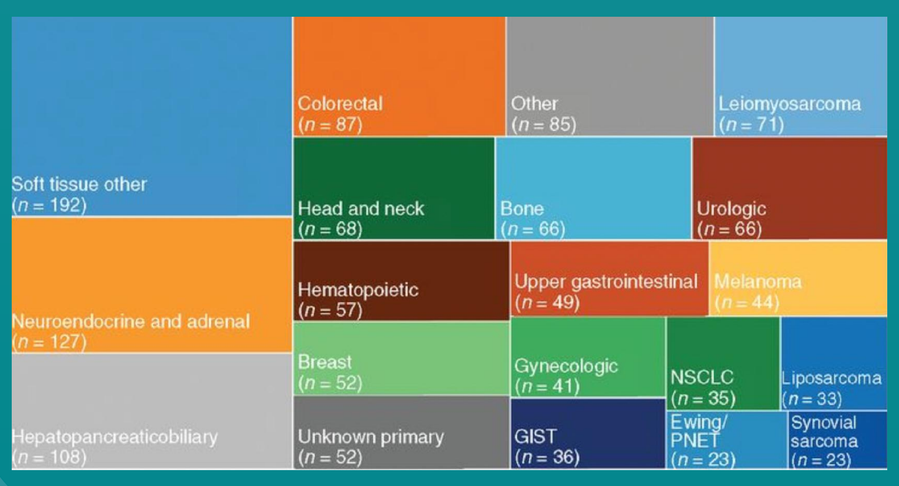
The Validator performs checks on submitted metadata to synchronize with the metadata model




The GHGA GitHub repository is regularly updated with latest changes pertaining to the model, schema and submission process

## Real world use case – MASTER dataset

The GHGA schema was used to capture metadata from Molecular Aided Stratification for Tumor Eradication (MASTER) – a large scale cohort study of rare cancers




1300 patients
6200 seq files
3 seq types
20 tumor types

MASTER ID: EGAS00001004813

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**References**

Wilkinson, Mark D., et al. "The FAIR Guiding Principles for scientific data management and stewardship." Scientific data 3.1 (2016): 1-9.
Horak, Peter, et al. "Comprehensive genomic and transcriptomic analysis for guiding therapeutic decisions in patients with rare cancers." Cancer discovery

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www.ghga-de.github.io/ghga-metadata-schema